ERCOFTAC
European Research Community On Flow, Turbulence And Combustion

3rd ERCOFTAC Workshop on Machine Learning for Fluid Dynamics

Conference Programme

(Last updated: 06/03/2026)

Conference Chairs:

- **Prof. Benjamin Sanderse –** *Centrum Wiskunde & Informatica, Eindhoven University of Technology*, chair
- **Dr. Richard P. Dwight –** *Delft University of Technology*, co-chair

Scientific Committee:

- **Prof. Ricardo Vinuesa –** *University of Michigan*
- **Prof. Nils Thuerey –** *Technical University of Munich*
- **Prof. Heng Xiao –** *University of Stuttgart*
- **Prof. Andrea Beck –** *University of Stuttgart*
- **Prof. Luca Biferale** – *University of Rome Tor Vergata*
- **Prof. Taraneh Sayadi –** *Conservatoire National Arts et Métiers*
- **Prof. Paola Cinnella –** *Sorbonne University*
- **Prof. Maria Vittoria Salvetti –** *University of Pisa*
- **Prof. Gianluca Iaccarino –** *Stanford University*
- **Dr. Romit Maulik -** *Pennsylvania State University*
- **Prof. Chris Pain –** *Imperial College London*
- **Dr. Jane Bae –** *California Institute of Technology*
- **Prof. Gianluigi Rozza –** *International School for Advanced Studies*
- **Prof. Elias Cueto –** *University of Zaragoza*
- **Prof. Angelo Iollo –** *University of Bordeaux*
- **Dr. Neil Ashton –** *NVIDIA*
- **Dr. Adrian Lozano Duran –** *California Institute of Technology*
- **Prof. Nathan Kutz –** *University of Washington*
- **Prof. Luca Magri –** *Imperial College London*
- **Dr. Daan Crommelin –** *Centrum Wiskunde & Informatica*
- **Prof. Gabriel Weymouth –** *Delft University of Technology*

Local Organizing Committee:

- **Dr. Nguyen Anh Khoa Doan –** *Imperial College London, Delft University of Technology*
- **Dr. Bernat Font –** *Delft University of Technology*
- **Dr. Wouter Edeling –** *Centrum Wiskunde & Informatica*
- **Tyler Buchanan –** *Delft University of Technology*
- **Nada Mitrovic –** *Centrum Wiskunde & Informatica*
- **Magdalena Yates –** *ERCOFTAC*

Program at a glance

Scientific Program	Wednesday 04 March
08:30 – 09:45	Arrival and registration
09:45 – 10:00	**Opening – Prof. Benjamin Sanderse** (Turing and streamed in Euler)
10:00 – 10:50	**Keynote 1 - *Causal Inference for Scientific Discovery in Fluid Dynamics*, Dr. Adrian Lozano Duran (CalTech)** (Turing and streamed in Euler)
10:50 – 11:30	Coffee break (Forum@CWI & Newton@CC)
11:30 – 12:30	Parallel sessions DR1 (Turing), EXP (Euler), O&D1 (L016-Hypatia), PIML 1 (L017-Ada)
12:30 – 13:40	Group picture + Lunch (Forum@CWI & Newton@CC)
13:40 – 15:20	Parallel sessions DR2 (Turing), TM-LES1 (Euler), O&D2 (L016-Hypatia), PIML 2 (L017-Ada)
15:20 – 16:00	Coffee break (Forum@CWI & Newton@CC)
16:00 – 17:40	Parallel sessions DR3 (Turing), TM-LES2 (Euler), ISI1 (L016-Hypatia), RFHT (L017-Ada)
Social Program	
17:45 – 19:15	Drinks reception (Forum@CWI & Newton@CC)

Scientific Program	Thursday 05 March
08:30 – 09:00	Arrival and registration
09:00 – 09:50	**Keynote 2 - *Solution discovery in fluids with high precision using neural networks*, Dr. Ching-Yao Lai (Stanford University)** (Turing and streamed in Euler)
09:50 – 10:30	Coffee break
10:30 – 12:10	Parallel sessions TM-RANS1 (Turing), DR4 (Euler), FI (L016-Hypatia), Dataset (L017-Ada)
12:10 – 13:20	Lunch (Forum@CWI & Newton@CC)
13:20 – 15:00	Parallel sessions TM-RANS2 (Turing), DR5 (Euler), Urban (L016-Hypatia), Surrogate1 (L017-Ada)
15:00 – 15:40	Coffee break (Forum@CWI & Newton@CC)
15:40 – 17:25	Parallel sessions Challenge (Turing), RL1 (Euler), Maritime (L016-Hypatia), Surrogate2 (L017-Ada)
Social Program	
19:00 –	Conference dinner (Tolhuistuin)

	Friday 06 March
08:30 – 09:00	Arrival and registration
09:00 – 09:50	**Keynote 3 - *Data over Dogma: Ruthless Empiricism, Strange Ideas, and the Future of Weather Forecasting*, Dr. Daniel Worrall (Google DeepMind)** (Turing and streamed in Euler)
09:50 – 10:30	Coffee break (Forum@CWI & Newton@CC)
10:30 – 12:10	Parallel sessions UQ+Q (Turing), RL2 (Euler), TM-RANS3 (L016-Hypatia), Surrogate3 (L017-Ada)
12:10 – 13:20	Lunch (Forum@CWI & Newton@CC)
13:20 – 15:00	Parallel sessions Solver (Turing), RL3 (Euler), Algo (L016-Hypatia)
15:00 – 15:30	Coffee break (Forum@CWI & Newton@CC)
15:30 – 16:20	**Keynote 4 – *Is AI what's next for fluids?*, Prof. Gianluca Iaccarino (Stanford University)** (Turing and streamed in Euler)
16:20 – 16:35	**Closure**

Euler Room Z.009
Newton Room Z.010
Turing Room Z.011
Entrance Congress Centre
Entrance CWI
MIVA
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L038
L039
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M043
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L036
L035
Paper Storage
M029
M028
Shower
Shower
L023
Stairs
lift
L022
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L020b
L020a
L020
L019
L018
Stairs
M020a
Reception
Stairs
M010
MIVA
WC
WC
L024
Scanner Studio
L028 Activity Room
Storage
pantry
WC
WC
L032
L033
WC
MIVA
Stairs
L009
L010
L011
L012
L013
L014
L015 Boardroom
L016 Hypatia Room
L017 Ada Room
lift

Detailed Program

Day 1 (Wednesday, 4th March 2026)

08:30 – 09:45 Arrival and registration
Room: Forum@CWI & Newton@CC

09:45 – 10:00 **Conference Opening**
Prof. Benjamin Sanderse (chair, Centrum Wiskunde & Informatica, Eindhoven University of Technology)
Room: Turing (also streamed in Euler)

10:00 – 10:50 **Keynote Talk:**
Causality is 4-D for Scientific Discovery in Fluids
Dr. Adrian Lozano Duran (California Institute of Technology)
Room: Turing (also streamed in Euler)

10:50 – 11:30 Coffee break
Room: Forum@CWI & Newton@CC

11:30 – 12:30 Parallel sessions (DR1, EXP, O&D1, PIML1)
Room: Turing, Euler, L016-Hypatia, L017-Ada

	Turing Dimensionality Reduction1 (DR1) Chair: Luca Magri	**Euler Experiments (EXP) Chair: David Rival**	**L016-Hypatia Optimization & Design1 (O&D1) Chair: Gianluca Iaccarino**	**L017-Ada Physics-informed ML1 (PIML1) Chair: Anh Khoa Doan**
11:30-11:50	*Data-driven Modeling of Chaotic Plane Couette flow on a Low-dimensional Spectral Submanifold* Kaszas Balint, Haller George	*Data driven rotor-wake dynamics: from measurements to sparse governing equations* Amico Enrico, Cafiero Gioacchino, Montagner Sara	*Deep Reinforcement Learning for Airfoil Shape Optimization: A Comparative Study of PPO and SAC* Bedri Yagiz, Lehmkuhl Oriol	*Enforcing Physics the Hard Way: On Constrained ML Architectures and Their Subtle Challenges in Fluid Dynamics* Cabral Manuel, Font Bernat, Weymouth Gabriel D.
11:50-12:10	*Stochastic Closure for Stable Long-Time Forecasting of Chaotic Systems* Cayuela Marco, Sayadi Taraneh, Le Chenadec Vincent, Schmid Peter	*Data-Driven Modelling of Transient Flyer Aerodynamics through Wind Tunnel Measurements* Wastell Owen, Marxen Olaf, Iacobello Giovanni	*AI-Machine Learning Methodologies to Accelerate CFD-based Design and Optimization for Turbomachinery Applications* Shahpar Shahrokh	*Conformal Physics-Informed Convolutional Neural Network for Reynolds-Averaged Navier Stokes Equations in Ducts* Bokil Gaurav, Merbold Sebastian, De Graaf Stefanie

12:10-12:30	*Non-intrusive oblique-projection-based model reduction with guaranteed-stable latent-space dynamics* Errico Cole, Padovan Alberto, Bodony Daniel	*Searching for a "textbook" of gust-wing encounters* Olivucci Paolo, Rival David E., Srivatsan Kowshik	*Machine Learning-based aerodynamic optimization of a low-Reynolds-number Outlet-Guide-Vane cascade* Li Shuai, Andersson Niklas	*Video-conditioned physics-informed neural networks for predicting hidden fluid mechanics in two-phase flows* Dreisbach Maximilian, Kiyani Elham, Kriegseis Jochen, Karniadakis George, Stroh Alexander

12:30 – 13:45 Group picture + Lunch
Room: Forum@CWI & Newton@CC

13:40 – 15:20 Parallel sessions (DR2, TM-LES1, O&D2, PIML2)
Room: Turing, Euler, L016-Hypatia, L017-Ada

	Turing Dimensionality Reduction2 (DR2) Chair: Andrea Novoa	**Euler Turbulence Modelling-LES1 (TM-LES1) Chair: Benjamin Sanderse**	**L016-Hypatia Optimization & Design2 (O&D2) Chair: Andrew Mole**	**L017-Ada Physics-informed ML2 (PIML2) Chair: Giovanni Stabile**
13:40-14:00	*A Latent Diffusion Transformer for Turbulence Generation and Data Assimilation* Teng Hao, Steinbrenner Fabian, Turan Baris, Xiao Heng	*Machine-learning wall model of large-eddy simulation across low- and high-speed flows over rough surfaces* Ma Rong, Lozano-Duran Adrian	*Fast LES-Based Geometry Optimization with Deep Reinforcement Learning and Solution Mapping* Sal Anglada Gaston, Lehmkuhl Oriol, Gomez Gonzales Samuel, Yagiz Bedri	*Weighting Physics and Data in PINNs: An Evaluation of Loss-Balancing Strategies for CFD* Correa Santiago, Díaz-Cuadro Christian, Draper Martín
14:00-14:20	*From Filtering to Forcing: Mimicking turbulence modeling via Data Assimilation and Machine Learning* Moussie Tom, Errante Paolo, Joesph Pierric, Meldi Marcello	**[ONLINE]** *Accelerated subgrid-scale modeling for passive scalar transport in LES of turbulent flows using data-driven methods* Sudhakar Surya Datta, Aditya Konduri, Lakkaraju Rajaram	*Efficient Thermo-Fluid System Optimization using Invariant CNNs and Parallel Active Learning* Koide Yuri, Kaithakkal Arjun J., Stroh Alexander, Friederich Pascal	*From Soft to Hard Turbulence: Architectural and Optimization Requirements of PINNs for Temperature Field Reconstruction in Rayleigh-Bénard Convection* Volk Marie-Christine, Mommert Michael, Bauer Christian, Sergent Anne, Lucor Didier, Wagner Claus
14:20-14:40	*Efficient Real-Time Adaptation of ROMs for Unsteady Flows Using Data Assimilation* Zighed Ismael, Novoa Andrea,	*Comparison of symmetry-preserving data-driven LES closures* Agdestein Syver Døving, Sanderse Benjamin	*Multi-Objective Wind Farm Bayesian Optimisation* Mole Andrew, Magri Luca	*Physics Informed Neural networks for downscaling on irregular meshes: reliability of convergence*

	Magri Luca, Sayadi Taraneh			Malhomme Nemo, Stabile Giovanni
14:40-15:00	*Latent-Space SINDy PDEs for Real-Time Digital Twins of Wall-Bounded Flows Perez* Cuadrado Miguel, Cavallazzi Giorgio Maria, Pinelli Alfredo	*Reinforcement-Learning-Based Wall Modeling for Turbulent Flows with Varying Pressure Gradients and Its Dependence on Subgrid-Scale Modeling* Bae Jane, Zhou Di	**WITHDRAWN** ~~*Multi-Agent Reinforcement Learning for Wind Farm Optimization in Atmospheric Boundary Layer*~~ ~~Manganelli Felice, Semeraro Onofrio, Leonardi Stefano, Cherubini Stefania, De Palma Pietro~~	*Sampling and weighting enhanced Physics Informed Neural Networks for highly turbulent Rayleigh-Bénard convection* Mrini Soufiane, Sergent Anne, Lucor Didier, Salort Julien, Chillà Francesca
15:00-15:20	*Real-time data assimilation in latent spaces for chaotic forecasting* Luca Magri, Özalp Elise, Novoa Andrea			*A Generalisable Normalisation Procedure for Physics-Informed Neural Network Training Data* Rawden Joshua, Vanderwel Christina, Symon Sean

15:20 – 16:00 Coffee break
Room: Forum@CWI & Newton@CC

16:00 – 17:40 Parallel sessions (DR3, TM-LES2, ISI, RFHT)
Room: Turing, Euler, L016-Hypatia, L017-Ada

	Turing Dimensionality Reduction3 (DR3) Chair: Taraneh Sayadi	**Euler Turbulence Modelling-LES2 (TM-LES2) Chair: Jane Bae**	**L016-Hypatia Inference, Sensor, and Inverse (ISI) Chair: Mahdi Abkar**	**L017-Ada Reacting Flow and Heat Transfer (RFHT) Chair: Salvatore Iavarone**
16:00-16:20	*Quantized local* reduced-order modelling. Colanera Antonio, Magri Luca	*Machine-learning-based subgrid-scale modeling and its application to flows over complex geometries* Ahnn Sangwoo, Kim Myunghwa, Choi Haecheon	*Smart Sensor Placement in Turbulent Flows: A Correlation-Aware Attribution Framework* Leung Sze Chai, Zhou Di, Bae Jane	*Machine Learning Tools for Studying Laser-Induced Ignition in a Rocket Combustor* Iaccarino Gianluca
16:20-16:40	*An explainable reduced-order model based on dynamic mode decomposition* Amor Christian, Corrochano Adrián, Le Clainche Soledad, Rosti Marco Edoardo	*Reduced Subgrid Scale Terms in Three-Dimensional Turbulence* Hoekstra Rik, Edeling Wouter	*Remember to "forget": overcoming the ill conditioned inverse problem in chaos by information theory* Fung Lloyd	*A data-driven reduced-order model for predicting flame dynamics via system identification* Novelli Chiara, Procacci Alberto, Parente Alessandro

16:40-17:00	*Bayesian Operator Inference for Galerkin projection closure and non-intrusive Reduced-Order Modelling* Procacci Alberto, Iavarone Salvatore	*Scalable Graph Network for Robust Sub-Grid Scale Modelling for Compressible Flows* Gonzalez Fernando, Larroque Anthony, Theret Aurelien, Drozda Dantas Martins Luciano, Odier Nicolas, Legaux Joeffrey, Teste Olivier	*Aerodynamic coefficients estimation from optimized surface pressure sensor distribution with multi-output Gaussian Processes Regression using wind tunnel data* Bucquet Quentin, Podvin Bérengère, Iavarone Salvatore, Hanna Badoui, Braud Caroline, Guilmineau Emmanuel	*Modelling chemistry integration via deep neural networks in LES-PDF simulations* of turbulent sooting flames Liu Weitao, Kronenburg Andreas, Zirwes Thorsten
17:00-17:20	*Proper Orthogonal Decomposition for initializing Auto-Encoders* Bousquet Rémi, Faucheu Alex, Nore Caroline, Lucor Didier	*Machine-learning-based largest-eddy simulation of turbulence and its generalisability to unseen conditions* Inubushi Masanobu, Satoshi Matsumoto, Susumu Goto	*Estimating Rotor-Effective Wind Speed Using Synthetic LiDAR–Based Wind-Field Reconstruction* Draper Martín, Schlipf David, Díaz-Cuadro Christian, Correa Santiago	*Physics-Informed Reduced-Order Modeling of Combustion Dynamics via Gaussian Process Regression and Data Assimilation* Lopez Constanza, Procacci Alberto, Podvin Bérengère, Iavarone Salvatore
17:20-17:40	*Reduced order model-based deep learning for flow prediction* Fiorini Camilla, Lepage Nicolas, Mortazavi Iraj, Beneddine Samir, Thome Nicolas, Sipp Denis			*Data-driven modeling of hypersonic flows in chemical non-equilibrium with catalytic surfaces* Sarras Konstantinos, Magin Thierry, Schmid Peter, Sayadi Taraneh

17:45 – 19:15 Drinks reception
Room: Forum@CWI & Newton@CC

Day 2 (Thursday, 5th March 2026)

08:30 – 09:00 Arrival and registration
Room: Forum@CWI & Newton@CC

09:00 – 09:50 **Keynote Talk:**
Solution discovery in fluids with high precision using neural networks
Dr. Ching-Yao Lai (Stanford University)
Room: Turing (also streamed in Euler)

09:50 – 10:30 Coffee break
Room: Forum@CWI & Newton@CC

10:30 – 12:10 Parallel sessions (TM-RANS1, DR4, FI, Dataset)
Room: Turing, Euler, L016-Hypatia, L017-Ada

	Turing **Turbulence Modelling-RANS1 (TM-RANS1)** **Chair: Ali Eidi**	**Euler** **Dimensionality Reduction4 (DR4)** **Chair: Angelo Iollo**	**L016-Hypatia** **Feature Identification (FI)** **Chair: Nils Thuerey**	**L017-Ada** **Dataset** **Chair: Ivette Rodriguez**
10:30-10:50	*CFD-driven symbolic regression of RANS transition models* Zemmour Louenas, Cinnella Paola, Gloerfelt Xavier	*Physics Informed Graph Neural Networks for Nonlinear Dynamics* Calandriello Gennaro, Romor Francesco, Stabile Giovanni	*Weak Dominant Balance: A robust method for identifying structure in more complex fluid flows* Ahnert Samuel, Lagemann Esther, Bae Jane, Vinuesa Ricardo, Lagemann Christian, Brunton Steven	*BCN16-HFWind: A High-Fidelity LES Dataset and its Application to Data-Driven Urban Flow Predictions* Duró Josep Maria, Mestres Ernest, Hernández Fabian, Calafell Joan, Lehmkuhl Oriol, Rodriguez Ivette
10:50-11:10	*Adjoint-Assisted CFD-Driven Symbolic RANS Modeling via Kolmogorov-Arnold Networks* Siddiqui Affan, Tian Renzhi, Dwight Richard	*A Reduced Order Model for predicting the dynamics of the Rayleigh-Taylor Instability* Granger Téo, Briard Antoine, Creusy Paul, Gréa Benoît-Joseph	*Feature Extraction from Complex Flow Fields: a Morphing-based Approach under Geometric Variability* Margheritti Riccardo, Semeraro Onofrio, Quadrio Maurizio, Boracchi Giacomo	*A Blended Wing Body Aircraft Dataset and Surrogate Model for Aerodynamic Predictions* Sung Nicholas, Spreizer Steven, Elrefaie Mohamed, Jones Matthew, Ahmed Faez
11:10-11:30	*Data-driven correction of a RANS-predicted turbulent heat flux applied to high-order CFD* Clément Levillain, Lodato Guido	*Low-dimensional representation of intermittent geophysical turbulence with statistics-informed convolutional autoencoders* Foldes Raffaello,	*Sweep events characterization through CNN* Saccaggi Enrico, Zaza Domenico, Iovieno Michele, Mendez Miguel Alfonso, Di Cicca Gaetano Maria	*Benchmarking State-of-the-Art Machine Learning Models on the DrivAerNet++ Dataset* Elrefaie Mohamed

		Camporeale Enrico, Marino Raffaele		
11:30-11:50	*Data-driven turbulence modelling in hypersonic flows using sparse symbolic regression* Calvi Paul, Gloerfelt Xavier, Breil Jérôme, Olazabal Marina, Cinnella Paola	*Reduced state space approach to identifying precursors to extreme events in turbulent flows* Kevin Schuurman, Dwight Richard, Doan Anh Khoa	*Predicting the linear instability eigenspectra using Deep Learning* Sánchez-Domínguez Miguel, De Vicente Javier, Lacasa Lucas, Chávez-Módena Miguel, Padilla-Montero Iván, Rodríguez Daniel, Valero Eusebio	*Identification of optimal training data for non-intrusive model reduction via operator inference* Rosenberger Henrik, Sanderse Benjamin, Stabile Giovanni
11:50-12:10	*Correcting Differential Reynolds Stress Models Using Data-Assimilation* Farro Gianmarco, Content Cédric, Volpiani Pedro, Sipp Denis	*A physics-based and theoretically proven precursor of extreme events with covariant Lyapunov vectors* Consonni Riccardo, Magri Luca	*Neural operator-based Global Stability and Resolvent Analysis* Wang Chengyun, Chen Liwei, Thuerey Nils	*Intelligent sampling for scientific model discovery: Learning to optimize and optimizing to learn* Larrañaga Ana, Fasel Urban, Brunton Steven L.

12:10 – 13:20 Lunch
Room: Forum@CWI & Newton@CC

13:20 – 15:00 Parallel sessions (TM-RANS2, DR5, Urban, Surrogate1)
Room: Turing, Euler, L016-Hypatia, L017-Ada

	Turing Turbulence Modelling-RANS2 (TM-RANS2) Chair: Heng Xiao	**Euler Dimensionality Reduction5 (DR5) Chair: Gianluigi Rozza**	**L016-Hypatia Urban Flow (Urban) Chair: Maria Vittoria Salvetti**	**L017-Ada Surrogate Modelling1 (Surrogate1) Chair: Antonio Colanera**
13:20 - 13:40	*Coupling data assimilation and machine learning to improve the Spalart Allmaras turbulence model for transonic flows* Carduner Louis, Fanizza Bartolomeo, Volpiani Pedro Stefanin, Renac Florent, Sipp Denis	*MOReDDPM - DDPMs for Non-intrusive Reduced-order Modeling of Parametric Fluid Flows* Nikken Michiel, Botteghi Nicolò, Califano Federico, Glas Silke	**[ONLINE]** *Urban Digital Twins at City Scale: Coupling High-Fidelity CFD with Uncertainty-Aware Machine Learning* Kokkinakis Ioannis, Christakis Nicholas, Drikakis Dimitris	*Discretization Invariant Neural Field Surrogates for Large-Scale Aerodynamic Simulation* Catalani Giovanni, Bertrand Xavier, Tost Frédéric, Bauerheim Michaël, Morlier Joseph
13:40 - 14:00	*Using Field Inversion Machine Learning to model the Reynolds Stress Tensor in the SU2 RANS flow solver* Boxho Margaux, Goffart Nicolas, Toulorge Thomas	*Optimal transport-based reduced order modeling for two-phase flows* Khamlich Moaad, Tonicello Niccolò, Pichi Federico, Rozza Gianluigi	*Meta-Learning Enhanced Physics-Informed Neural Networks for Predicting Urban Canyon Wind Flows* Antoniou Nestoras, Laakkonen Janne, Mouzourides Petros,	*Towards foundation surrogate model for flow fields around wings* Yang Yunjia, Thuerey Nils

			Neophytou Marina, Hautamäki Ville	
14:00 - 14:20	*Data-Driven Correction of RANS Models for Urban Flow Prediction via Data Assimilation and Machine Learning* Rosellini Matteo, Ursetto Aurora, Tavazzi Pietro, Mariotti Alessandro, Stabile Giovanni, Salvetti Maria Vittoria	*Latent-Space Model Predictive Control for Partially-Observable High-Dimensional Systems* Marra Luigi, Semeraro Onofrio, Mathelin Lionel, Meilán-Vila Andrea, Discetti Stefano	*Towards predicting urban flows with GAVI and SHRED* Eximeno Benet, Miró Arnau, Duró Josep Maria, Kutz J. Nathan, Lehmkuhl Oriol, Rodriguez Ivette	*Lossless Domain Mapping for Learning CFD Fields across Geometries* Tamburini Emanuele, Cinnella Paola, Belme Anca-Claudia
14:20 - 14:40	*Bayesian inversion of RANS turbulence models in Flow-MRI* Namuroy Claire, Kontogiannis Alexandros, Juniper M.P.	*Predictive reduced order models of unsteady chaotic flows using transformer architecture* Taneja Kashish, Sayadi Taraneh, Mortazavi Iraj	*Neural Physics: Using machine-learning–libraries to build a differentiable physics-based solver applied to urban flows* Heaney Claire, Chen Boyang, Pain Christopher	*Machine-Learning-Based Surrogate Modeling for Stochastic Multiscale Simulation Methodology* Kim Changho, Xu Zihan, Nonaka Andrew, Zhu Yuanran
14:40 - 15:00	*Field-Inversion and Machine-Learning–Assisted Correction of the Compressible RANS Equations: Application to Shock-Wave/Turbulent Boundary-Layer Interactions* Fanizza Bartolomeo, Volpiani Pedro Stefanin, Renac Florent, Sipp Denis	*Generalization capabilities of Transformers in fluid dynamics* Ifaistos Theofanis, Nastorg Matthieu, Menier Emmanuel, Charpiat Guillaume, Schoenauer Marc	*Reflection-Equivariant Graph Neural Networks for Predicting Wind Loads on Buildings* Vargiemezis Themistoklis, Kanatsoulis Charilaos, Gorle Catherine	

15:00 – 15:40 Coffee break
Room: Forum@CWI & Newton@CC

15:40 – 17:25 Parallel sessions (Challenge, RL1, Maritime, Surrogate2)
Room: Turing, Euler, L016-Hypatia, L017-Ada

	Turing Data Challenge (Challenge) Chair: Tyler Buchanan		**Euler Reinforcement Learning1 (RL1) Chair: Anh Khoa Doan**	**L016-Hypatia Maritime application (Maritime) Chair: Bernat Font**	**L017-Ada Surrogate Modelling2 (Surrogate2) Chair: Didier Lucor**
15:40-16:00	*CYPHER machine learning challenge 2025: a community effort for data-driven turbulent combustion modeling* Lorenzo Piu, Lapenna Pasquale, Bottari Stefano, Osseily Tamara, Baffetti Tommaso, Indelicato Giuseppe, Shigematsu Kosuke, Tocilla Albina, Kazani Jonid, Attili Antonio, Parente Alessandro	15:40-16:00	*Controlling Mode Transitions in Rotating Detonation Engines using Deep Reinforcement Learning* Holme Kristian, Rabault Jean, Vinuesa Ricardo, Mortensen Mikael	*Machine Learning approaches to predict shape sensitivities for ship hulls* Arian Maram Moloud, Bletsos Georgios, Schwarz Henning, Palm Michael, Rung Thomas	*IGM: A physics-informed deep learning model for large-scale ice-sheet flow* Gregov Thomas, Rosier Sebastian, Finley Brandon, Vieli Andreas, Jouvet Guillaume
16:00-16:10	*FluidsBench: A benchmark for the next generation of Computational Fluid Dynamics AI models* Neil Ashton	16:00-16:20	**[ONLINE]** *Deep Reinforcement Learning for Active Control of Vortex-Induced Vibrations* Alhussein Hussam, Font Bernat, Daqaq Mohammed	*Predicting pressure distributions on surface ship hulls using boundary graph neural networks* Jena Sankalp, Weymouth Gabriel, Lidtke Artur, Coraddu Andrea	*Downscaling simulation dynamics for regional weather forecasting* Jackaman James, Celledoni Elena
16:10-16:25	*Presentation of the Turbulence Modelling Challenge* Tyler Buchanan	16:20-16:40	*Deep reinforcement learning for heat transfer control in turbulent convection* Zhou Zisong, Zhu Xiaojue	*Prediction of ship hydrodynamic resistance with artificial neural networks* Lefkiou Christoforos, Koukouvinis Phoevos (Foivos), Chatzis Sotirios	*Proof of Concept of a Real-Time Fluid Dynamics Model for Immersive Environments Using Artificial Intelligence* Laot Titouan, Marchal Antoine, Perrot Vincent
16:25-16:40	*Towards a machine learning-augmented one-equation turbulence model for external aerodynamics in open-source framework* Di Fabbio Tony, Segalerba Eric, Guerrero Rivas Joel Enrique	16:40-17:00	*Reinforcement Learning for Complex Flows Modelling* Helal Isa, Giauque Alexis, Corre Christophe, Sauret Emilie		
16:40-16:55	*Data-Driven RANS Closures Using Adaptive Curvature Classifier for Vortex Flows* Buchanan Tyler,				

	West Alastair, Dwight Richard				
16:55- 17:10	*Symbolic Turbulence Modelling with Multi-Agent Reinforcement Learning* Tan Zing Shawn, Eidi Ali, Buchanan Tyler, Dwight Richard P.				
17:10- 17:25	*Towards a unified turbulence model through multi-objective learning* Wang Haochen, Liu Zhuoran, Zhao Zhuolin, Xiao Heng				

19:00 – Conference dinner (walk-in from 19:00, dinner starts at 19:30)
Tolhuistuin, IJpromenade 2, 1031 KT Amsterdam
https://maps.app.goo.gl/qaDZKEc9fTU6NnpC8

Day 3 (Friday, 6th March 2026)

08:30 – 09:00 Arrival and registration
Room: Forum@CWI & Newton@CC

09:00 – 09:50 **Keynote Talk:**
Data over Dogma: Ruthless Empiricism, Strange Ideas, and the Future of Weather Forecasting
Dr. Daniel Worrall (Google DeepMind)
Room: Turing (also streamed in Euler)

09:50 – 10:30 Coffee break
Room: Forum@CWI & Newton@CC

10:30 – 12:10 Parallel sessions (UQ+Q, RL2, TM-RANS3, Surrogate3)
Room: Turing, Euler, L016-Hypatia, L017-Ada

	Turing Uncertainty Quantification and Quantum (UQ+Q) Chair: Richard Dwight	**Euler Reinforcement Learning2 (RL2) Chair: Kamila Zdybał**	**L016-Hypatia Turbulence Modelling-RANS3 (TM-RANS3) Chair: Paola Cinnella**	**L017-Ada Surrogate Modelling3 (Surrogate3) Chair: Wouter Edeling**
10:30-10:50	*A Surrogate-Informed Sparse Grid Interpolation for Uncertainty Quantification and Sensitivity Analysis of Complex Flows* Salvetti Maria Vittoria, Rosellini Matteo, Fruzza Filippo, Mariotti Alessandro, Tamellini Lorenzo	*Viscoelastic Turbulent Channel Drag Reduction via Opposition Control Using Multi-Agent Reinforcement Learning* Sharma Udit, Beneitez Miguel, Wittberg Lisa Prahl, Mirjalili Shahab, Vinuesa Ricardo, Tammisola Outi	*Development of a RANS Local Pressure-Gradient Closure for Transition Modeling* Capel Jorquera Javier, Cardesa José Ignacio, Pascal Lucas, Chávez-Módena Miguel, Valero Eusebio, González Leo Miguel	*Toward generative surrogate models of hydrodynamic instabilities and turbulent mixing* Thévenin Sébastien, Sterbentz Dane, Korner Kevin, Haberstich Cécile, Briard Antoine, Gréa Benoît-Joseph, Nadiga Balasubramanya T., Schill William, Belof Jonathan L.
10:50-11:10	*Shape generative models with LDDM's flow conditional matching: an application to uncertainty quantification in hemodynamics* Romor Francesco, Caiazzo Alfonso, Katz Sarah	*Wall-to-Flow: GAN-Assisted Deep Reinforcement Learning for Turbulent Flow Control from Sparse Wall Measurements* Cavallazzi Giorgio Maria, Perez Cuadrado Miguel, Pinelli Alfredo	*Generalizable Graph Neural Operators for Field Inversion Based RANS Corrections* Stumpo Leonardo, Ferrero Andrea, Masseni Filippo, Pastrone Dario	*Flow-based generative modeling of turbulent channel flow fields* Aerts Frederik, Nuyens Dirk, Meyers Johan
11:10-11:30	*Shape-informed surrogate modeling and applications to data assimilation and graph neural networks* Caiazzo Alfonso,	*Reinforcement learning for autonomous observation in optical velocimetry* Zdybał Kamila,	*Multi-Fidelity Data-Driven Modeling, Reynolds Number Extrapolation, and a log(Re) Solution to the Turbulence Problem*	*Temporal coherency in high-resolution 3D neural PDE surrogates* Luca Guastoni, Thuerey Nils

	Romor Francesco, Goubergrits Leonid, Galarce Felipe, Brüning Jan	Mucignat Claudio, Lunati Ivan	Li Jiaqi, Huang George, Kunz Robert, Yang Xiang	
11:30-11:50	*Solving nonlinear differential equations with quantum computers and a Fokker-Planck embedding* Ahmed Osama, Tennie Felix, Magri Luca		*Data-Driven Turbulence Modeling using Coarse-Grid Nonlinear Super-Stencils with Wall Functions* Lyu Shiyu, Wang Zhong-Nan	*Neural Operators For Accelerating Flow Field Predictions in a 2D Compressor Cascade* Xavier Alan, Renson Ludovic
11:50-12:10	*Learning a Surrogate Quantum Circuit for the Collision Operator in the Lattice Boltzmann Method* Lacatus Monica, Möller Matthias		*Learning a Heat-Flux Closure for Jet-in-Crossflow Using Indirect Data* Zhao Zhuolin, Wang Haochen, Xiao Heng	*ML-based correction of coarse CFD simulations for turbomachinery flows* Tschisgale Silvio, Eichenhardt Timo, Kempe Tobias

12:10 – 13:20 Lunch
Room: Forum@CWI & Newton@CC

13:20 – 15:00 Parallel sessions (Solver, RL3, Algo)
Room: Turing, Euler, L016-Hypatia

	Turing Solver Chair: Alexis Giauque	**Euler Reinforcement Learning3 (RL3) Chair: Saeed Salehi**	**L016-Hypatia Algorithms (Algo) Chair: Luca Guastoni**
13:20-13:40	*Accelerating Numerical Simulations in CFD by Model Reduction and Scientific Machine Learning* Rozza Gianluigi	*SINDy-RL: Interpretable and Efficient Model-Based Reinforcement Learning* Zolman Nicholas, Lagemann Christian, Fasel Urban, Kutz J. Nathan, Brunton Steven L.	*Gaussian Artificial Entropy viscosity via machine Learning sensor* Limonchi Samuel, Rodriguez Ivette, Lehmkuhl Oriol
13:40-14:00	*Acceleration of an AMG pressure solver using graph neural networks* Eric Chillon, Lidtke Artur, Doan Nguyen Anh Khoa, Font Bernat	*Improvement turbulence control through explainable deep learning* Beneitez Miguel, Cremades Andrés, Guastoni Luca, Vinuesa Ricardo	*An octree-based sampling algorithm for processing big simulation data* Geise Janis, Spinner Sebastian, Weiner Andre
14:00-14:20	*diffSPH: Differentiable Smoothed Particle Hydrodynamics for Adjoint Optimization and Machine Learning* Winchenbach Rene, Thuerey Nils	*Transfer learning strategies for accelerating reinforcement-learning-based flow control* Saeed Salehi	*Stabilizing PDE-ML systems with applications to fluid dynamics* Stinis Panos, Howard Amanda, Ahmed Shady, Qadeer Saad

14:20-14:40	*Interpretable and differentiable fluid flow solvers in latent spaces with operator learning* Arzani Amirhossein	*Gradient-enhanced reinforcement learning for fluid flow control using differentiable simulations* Mokbel Sajeda, Lagemann Christian, Lagemann Esther, Brunton Steven	
14:40-15:00	*Bridging Physics and Learning: End-to-End and Residual-Based Optimization in Differentiable Solvers* Saverio Luca, Bucci Michele Alessandro, Content Cédric, Sipp Denis		

15:00 – 15:30 Coffee break
Room: Forum@CWI & Newton@CC

15:30 – 16:20 **Keynote Talk:**
Is AI what's next for fluids?
Prof. Gianluca Iaccarino (Stanford University)
Room: Turing (also streamed in Euler)

16:20 – 16:35 **Closing remark**